A Monte Carlo Evaluation of Tests for Comparing Dependent Correlations

JAMES B. HITTNER
KIM MAY
Department of Psychology
College of Charleston

N. CLAYTON SILVER
Department of Psychology
University of Nevada, Las Vegas

ABSTRACT. The authors conducted a Monte Carlo simulation of 8 statistical tests for comparing dependent zero-order correlations. In particular, they evaluated the Type I error rates and power of a number of test statistics for sample sizes (*N*s) of 20, 50, 100, and 300 under 3 different population distributions (normal, uniform, and exponential). For the Type I error rate analyses, the authors evaluated 3 different magnitudes of the predictor–criterion correlations ($\rho_{y,x1} = \rho_{y,x2}$ = .1, .4, and .7). For the power analyses, they examined 3 different effect sizes or magnitudes of discrepancy between $\rho_{y,x1}$ and $\rho_{y,x2}$ (values of .1, .3, and .6). They conducted all of the simulations at 3 different levels of predictor intercorrelation ($\rho_{x1,x2}$ = .1, .3, and .6). The results indicated that both Type I error rate and power depend not only on sample size and population distribution, but also on (a) the predictor intercorrelation and (b) the effect size (for power) or the magnitude of the predictor–criterion correlations (for Type I error rate). When the authors considered Type I error rate and power simultaneously, the findings suggested that O. J. Dunn and V. A. Clark's (1969) *z* and E. J. Williams's (1959) *t* have the best overall statistical properties. The findings extend and refine previous simulation research and as such, should have greater utility for applied researchers.

Key words: dependent correlations, Monte Carlo simulation, statistical power, Type I error rate, zero-order

WHEN TWO OR MORE CORRELATION COEFFICIENTS are calculated from a single sample, they are regarded as dependent. Consider, for example, a situation in which a researcher is interested in whether frequency of alcohol use (y) is more highly positively correlated with depression (x_1) or anxiety (x_2) within a

Address correspondence to James B. Hittner, Department of Psychology, College of Charleston, Charleston, SC 29424; hittnerj@cofc.edu (e-mail).

sample of individuals attending an outpatient alcohol treatment program. In that example, the two sample correlations of interest are $r_{y,x1}$ and $r_{y,x2}$ and the null hypothesis to be tested is H_0: $\rho_{y,x1} = \rho_{y,x2}$. To test that null hypothesis, researchers have proposed a number of test statistics, most of which purport to follow either the t distribution or the standard normal distribution. Moreover, the Type I error rates and statistical power of many of those tests have been evaluated via Monte Carlo simulation.

In an early simulation study, Neill and Dunn (1975) examined the Type I error rates and power of 11 different tests for comparing dependent correlations. Their results indicated that for small- to moderate-sized samples (*N*s of 10, 26 and 50), Williams's (1959) modification of Hotelling's (1940) t performed best in terms of both Type I error rate and power. Williams's t also performed well when the sampling distribution of r was highly skewed. In addition to Williams's t, Dunn and Clark's (1969) z test also effectively controlled Type I error rate while it maintained adequate power. Although Hotelling's t was consistently more powerful than was Williams's t, the Type I error rates for Hotelling's test were uniformly unsatisfactory (Neill & Dunn). Steiger (1980) also noted that Hotelling's t was overly powerful at the expense of Type I error rate. As with Neill and Dunn, Boyer, Palachek, and Schucany (1983) found that of the five statistics that they examined in their simulation, Williams's t generally showed the best Type I error rate control and power. However, unlike Neill and Dunn, Boyer et al. found that the performance of Williams's t was compromised when they used a highly skewed distribution (the lognormal).

May and Hittner (1997) conducted a simulation study of Williams's t, Hotelling's t, Olkin's (1967) z test, and Meng, Rosenthal, and Rubin's (1992) z test to examine further the performance of those procedures. However, it is important to note that the version of Williams's t that May and Hittner simulated, and which was originally published in Hendrickson, Stanley, and Hills (1970), is mathematically different from the Williams's t that has been reported by others (e.g., Neill & Dunn, 1975; Steiger, 1980). May and Hittner's results indicated that relative to the z tests, the t tests maintained statistical power at the expense of inflated Type I error rate. In addition, the Type I error rates and power of Hendrickson et al.'s version of Williams's t were quite similar to those of Hotelling's t. A strength of May and Hittner's study is that in addition to evaluating each test statistic in the context of multiple sample sizes and different population distributions (the standard Monte Carlo procedure), they also systematically varied (a) the magnitude of discrepancy between $\rho_{y,x1}$ and $\rho_{y,x2}$ (for power), (b) the magnitude of the equal values of $\rho_{y,x1}$ and $\rho_{y,x2}$ (for Type I error rate), and (c) the magnitude of the predictor intercorrelation, $\rho_{x1,x2}$. May and Hittner's results indicated that the power and the Type I error rate of a given test statistic are determined by the combined effects of the aforementioned factors.

Although the methodology used by May and Hittner (1997) appears to be an improvement over previous approaches, there are several ways in which their study could have been strengthened. First, May and Hittner did not simulate the version of Williams's t (i.e., the "standard") that has been examined in previous studies (e.g., Boyer et al., 1983; Neill & Dunn, 1975). Second, there are several additional z tests that have received attention in the literature but were not simulated by May and Hittner. In particular, the empirical results of Neill and Dunn suggested that Dunn and Clark's z test might hold promise. Steiger (1980) likewise advocated the use of Dunn and Clark's z. He also noted in his review that Dunn and Clark's z statistic could likely be improved if one uses the arithmetic average of $r_{y,x1}$ and $r_{y,x2}$ rather than the individual correlations when one calculates the sample covariance term. Steiger maintained that his modification of Dunn and Clark's z should result in greater control over Type I error rate. One might also be able to obtain the average of $r_{y,x1}$ and $r_{y,x2}$ by using the backtransformed average Fisher's (1921) z. Silver and Dunlap (1987) have shown via Monte Carlo simulation that the backtransformed average z is generally less biased than is the average r when the sample size is small (i.e., Ns of 10, 20 and 30), and that average z becomes increasingly less biased for large values of average r (i.e., ≈ .5 and above). To the best of our knowledge, neither Steiger's modification of Dunn and Clark's z nor Steiger's modification using a backtransformed average Fisher's z procedure has been examined via Monte Carlo simulation.

In light of the aforementioned considerations and building on the work of May and Hittner (1997), our purpose in the present Monte Carlo study was to evaluate comparatively the Type I error rates and power of the following eight statistical tests: Hotelling's t; Williams's t per Hendrickson et al.; Olkin's z; Meng et al.'s z; Williams's standard t; Dunn and Clark's z; Steiger's modification of Dunn and Clark's z; and Steiger's modification of Dunn and Clark's z with a backtransformed average Fisher's z procedure. May and Hittner studied the first four of these tests, but they did not examine the last four. As such, the present study allowed for a more comprehensive and integrative evaluation of statistical tests for comparing dependent correlations. Table 1 presents the formulas for all eight statistics.

Method

The simulation procedure that we used in the present study was identical to that used in the study by May and Hittner (1997). We reiterate the procedure here for the benefit of interested researchers and for clarity of presentation. Suppose that we have observations on a criterion (y) and two predictors (x_1 and x_2). Then, for a test of H_o: $\rho_{y,x1} = \rho_{y,x2}$, the population correlation matrix $\mathbf{R}$ will be a 3 × 3 matrix containing $\rho_{y,x1}$, $\rho_{y,x2}$, and $\rho_{x1,x2}$. For each specified 3 × 3 correlation matrix $\mathbf{R}$, the Cholesky decomposition $\mathbf{U}'\mathbf{U} = \mathbf{R}$ was computed, where $\mathbf{U}$ is the matrix of factor loadings on the principal components for the square-root method of factoring a correlation matrix, and $\mathbf{U}'$ is the transpose of $\mathbf{U}$. Then, for N observations

TABLE 1. Formulas for the Eight Test Statistics

Hotelling's (1940) t statistic

$$t = \frac{(r_{yx1} - r_{yx2})\sqrt{(N-3)(1+r_{x1x2})}}{\sqrt{2(1 + 2r_{yx1}r_{yx2}r_{x1x2} - r_{yx1}^2 - r_{yx2}^2 - r_{x1x2}^2)}}.$$

Williams' (1959) "standard" t statistic

$$t = \frac{(r_{yx1} - r_{yx2})}{\sqrt{\dfrac{(N-1)(1+r_{x1x2})}{2\left[\left(\dfrac{N-1}{N-3}\right)|R| - \bar{r}^2(1-r_{x1x2})^3\right]}}}.$$

Williams' modified t per Hendrickson, Stanley, and Hills (1970)

$$t = \frac{(r_{yx1} - r_{yx2})\sqrt{(N-3)(1+r_{x1x2})}}{\sqrt{2|R| + \dfrac{(r_{yx1} - r_{yx2})^2(1-r_{x1x2})^3}{4(N-1)}}}.$$

Olkin's (1967) z statistic

$$z = \frac{\sqrt{N}(r_{yx1} - r_{yx2})}{\sqrt{(1-r_{yx1}^2)^2 + (1-r_{yx2}^2)^2 - 2r_{x1x2}^3 - (2r_{x1x2} - r_{yx1}r_{yx2})(1 - r_{yx1}^2 - r_{yx2}^2 - r^2{}_{x1x2})}}.$$

Meng, Rosenthal, and Rubin's (1992) z statistic

$$z = (z_{ryx1} - z_{ryx2})\sqrt{\frac{N-3}{2(1-r_{x1x2})h}},$$

where

$$h = \frac{1 - f\bar{r}^2}{1 - \bar{r}^2} = 1 + \frac{\bar{r}^2}{1-\bar{r}^2}(1-f)$$

and

$$f = \frac{1 - r_{x1x2}}{2(1-\bar{r}^2)}, \text{ which must be } \leq 1.$$

(table continues)

TABLE 1. (Continued)

Dunn and Clark's (1969) z statistic

$$z = \sqrt{N-3}(z_{ryx1} - z_{ryx2})(2 - 2\,\mathrm{cov}_{yx1x2})^{-1/2},$$

where

$$\mathrm{cov}_{yx1x2} = \frac{r_{x1x2}(1 - r_{yx1}^2 - r_{yx2}^2) - .5(r_{yx1}r_{yx2})(1 - r_{yx1}^2 - r_{yx2}^2 - r_{x1x2}^2)}{(1 - r_{yx1}^2)(1 - r_{yx2}^2)}.$$

Steiger's modification of Dunn and Clark's z

$$z = \sqrt{N-3}(z_{ryx1} - z_{ryx2})(2 - 2\,\overline{\mathrm{cov}}_{yx1x2})^{-1/2},$$

where

$$\overline{\mathrm{cov}}_{yx1x2} = \frac{r_{x1x2}(1 - \bar{r}^2 - \bar{r}^2) - .5(\bar{r}^2[1 - \bar{r}^2 - \bar{r}^2 - r_{x1x2}^2])}{(1 - \bar{r}^2)^2}.$$

Steiger's modification of Dunn and Clark's z using a backtransformed average Fisher's z procedure

$$z = \sqrt{N-3}(z_{ryx1} - z_{ryx2})(2 - 2z\,\mathrm{cov}_{yx1x2})^{-1/2},$$

where

$$r_{\bar{z}} = \frac{e^{2\bar{z}-1}}{e^{2\bar{z}+1}}$$

and

$$z\,\mathrm{cov}_{yx1x2} = \frac{r_{x1x2}(1 - r_{\bar{z}}^2 - r_{\bar{z}}^2) - .5(r_{\bar{z}}^2)(1 - r_{\bar{z}}^2 - r_{\bar{z}}^2 - r_{x1x2}^2)}{(1 - r_{\bar{z}}^2)^2}.$$

and three variables (the criterion and two predictors), an $N \times 3$ matrix of uncorrelated observations **X** was generated from a specified distribution (normal, uniform, or exponential). The calculation of the matrix product **XU** then produces an $N \times 3$ matrix of observations that have, in the population, the correlation matrix specified in **R** (see Browne, 1968). For each specified correlation matrix, we repeatedly drew 2,000 samples of size N and obtained empirical estimates of Type I error rate (α) and power for Hotelling's t, Williams's t per Hendrickson et al., Olkin's z, Meng et al.'s z, Williams's standard t, Dunn and Clark's z, Steiger's modification of Dunn and Clark's z, and Steiger's modification of Dunn and Clark's z using a backtransformed average Fisher's z procedure. For each test statistic, estimates of Type I error rate and power were obtained under three different specified population distributions (normal, uniform, and exponential).

For both sets of analyses (i.e., Type I error rates and power), we used sample sizes of 20, 50, 100, and 300 and predictor intercorrelations ($\rho_{x1.x2}$) of .1, .3, and .6. For the power analyses, we examined the extent to which different degrees of discrepancy between $\rho_{y,x1}$ and $\rho_{y,x2}$ affected the results. Because we varied the magnitude of discrepancy between $\rho_{y,x1}$ and $\rho_{y,x2}$ (rather than examining the impact of a single fixed discrepancy as is typically done), our power simulations should be more useful to the applied researcher who, when working with empirical data, often encounters sample predictor–criterion correlations of varying magnitudes that differ in degree from each other. The magnitudes of discrepancy between $\rho_{y,x1}$ and $\rho_{y,x2}$ (hereinafter referred to as *effect sizes*) that we examined were .1, .3, and .6. For the Type I error rate analyses, which examined the proportion of cases in which the null hypothesis was rejected when the two population correlations were equal, we evaluated the values for the two population correlations ($\rho_{y,x1}$ and $\rho_{y,x2}$) of .1, .4, and .7.

Our selection of three different population correlation values for examining Type I error rates should also be of greater use to applied researchers. We performed all of the Monte Carlo simulations using SAS PROC IML and SAS data step language (SAS Institute, 1990)

Results

Type I Error Rate

The Type I error rate simulation results are presented in Table 2. Given the 2,000 iterations, the empirical alphas that were near the nominal level of .05 (i.e., αs ranging from .04 to .06) had standard errors that ranged from ± .0044 to ± .0053. Given the relatively small magnitudes of the standard errors, it appears that the influence of sampling variability on the empirical alphas was minimal. There were considerable differences in Type I error rates across the eight test statistics for the various sample sizes, effect sizes, and population distributions, as the values in Table 2 indicate. Nevertheless, there were some trends that emerged despite the differences. For example, across all sample sizes and for each population distribution, Hotelling's t and Hendrickson et al.'s modification of Williams's t had substantially greater Type I error rates than did the other six statistics when the predictor–criterion correlation was large (.7) and the predictor intercorrelations were small to moderate (.1 and .3). For example, consider the normal distribution situation in which the predictor–criterion correlation is .7, the predictor intercorrelation is .1, and the sample size is 20. In that case, Hotelling's t and Hendrickson et al.'s modification of Williams's t yielded Type I error rates of .148, whereas the other statistics yielded values ranging from .042 to .052. Moreover, Type I error rates were above the nominal level for Olkin's z test, given the smallest sample size ($N = 20$) and small and medium (.1 and .4) predictor–criterion correlations for all three distributions. Although the Meng et al. procedure exhibited reasonable control of Type

TABLE 2. Type I Error Rates for all Eight Statistics (Nominal α = .05)

Pred–crit. corr.	Pred. int.	H	WH	O	M	WS	D	S1	S2
				Normal (N = 20)					
.1	.1	048	048	068	046	043	046	042	042
	.3	050	050	065	045	059	060	058	058
	.6	048	048	064	044	054	053	051	051
.4	.1	050	049	064	044	040	042	040	040
	.3	052	052	062	044	048	049	048	048
	.6	052	052	066	050	044	044	042	042
.7	.1	148	148	042	045	046	050	048	052
	.3	072	072	042	044	047	049	049	048
	.6	057	056	048	046	040	042	042	040
				Uniform (N = 20)					
.1	.1	054	054	074	048	052	055	053	053
	.3	054	054	072	049	047	048	046	046
	.6	053	053	068	047	051	052	049	049
.4	.1	054	054	064	046	047	049	048	047
	.3	052	052	060	044	048	051	049	048
	.6	046	046	055	040	049	050	048	048
.7	.1	152	151	042	043	042	048	044	050
	.3	072	072	042	044	036	040	038	040
	.6	038	038	031	035	029	032	031	030
				Exponential (N = 20)					
.1	.1	050	050	074	048	049	051	050	050
	.3	046	046	068	042	050	050	049	049
	.6	042	042	056	037	044	044	040	040
.4	.1	080	080	089	068	074	077	076	076
	.3	072	072	082	064	066	070	068	066
	.6	060	060	068	055	066	068	064	062
.7	.1	182	182	086	091	079	082	081	082
	.3	128	128	094	102	092	102	100	098
	.6	117	116	102	109	107	112	110	108
				Normal (N = 50)					
.1	.1	054	054	063	054	047	048	047	047
	.3	058	058	066	058	048	048	048	047
	.6	065	065	070	063	050	050	050	049

(table continues)

TABLE 2. (Continued)

Pred–crit. corr.	Pred. int.	H	WH	O	M	WS	D	S1	S2
		Normal (N = 50)							
.4	.1	062	062	058	050	046	046	046	046
	.3	058	058	062	052	052	054	053	052
	.6	060	060	066	058	049	050	049	049
.7	.1	148	148	045	046	050	051	050	052
	.3	080	080	050	050	053	054	054	054
	.6	064	064	054	056	048	050	049	048
		Uniform (N = 50)							
.1	.1	044	044	052	043	045	046	045	045
	.3	046	046	055	044	046	047	046	046
	.6	049	049	055	048	047	048	047	047
.4	.1	046	046	044	037	050	052	051	050
	.3	042	042	044	040	043	044	043	043
	.6	044	044	047	043	038	040	038	036
.7	.1	156	156	040	042	044	044	044	044
	.3	064	064	035	036	038	040	040	040
	.6	034	034	029	030	026	028	028	026
		Exponential (N = 50)							
.1	.1	046	046	052	044	053	053	053	053
	.3	044	044	047	042	038	038	038	037
	.6	035	035	042	033	040	040	039	039
.4	.1	087	087	080	074	066	068	066	066
	.3	075	075	076	073	068	070	068	068
	.6	068	068	069	068	065	065	064	064
.7	.1	189	189	081	082	064	064	064	066
	.3	140	140	100	102	100	100	100	100
	.6	119	119	111	113	106	106	106	106
		Normal (N = 100)							
.1	.1	058	058	060	056	048	048	048	048
	.3	058	058	062	058	045	045	045	045
	.6	054	054	058	053	054	054	054	054
.4	.1	060	060	058	054	048	048	048	048
	.3	062	062	061	057	050	050	050	050
	.6	060	060	061	058	051	052	051	051

(table continues)

TABLE 2. (Continued)

Pred–crit. corr.	Pred. int.	H	WH	O	M	WS	D	S1	S2
				Normal (N = 100)					
.7	.1	170	170	050	051	052	052	052	052
	.3	084	084	054	054	047	048	047	047
	.6	062	062	056	056	051	051	051	051
				Uniform (N = 100)					
.1	.1	047	047	047	046	056	056	056	056
	.3	044	044	046	044	046	046	046	046
	.6	040	040	042	040	040	040	040	040
.4	.1	048	048	045	044	044	044	044	044
	.3	042	042	041	039	041	041	041	041
	.6	037	037	038	035	048	048	048	048
.7	.1	144	144	042	042	050	050	050	050
	.3	062	062	038	038	032	032	032	032
	.6	032	032	030	030	028	029	028	028
				Exponential (N = 100)					
.1	.1	048	048	050	047	050	051	050	050
	.3	042	042	046	040	047	047	047	047
	.6	042	042	045	042	049	049	049	049
.4	.1	070	070	066	066	061	061	061	061
	.3	070	070	066	064	062	062	062	062
	.6	064	064	064	063	068	068	068	068
.7	.1	177	177	074	074	062	064	064	064
	.3	130	130	095	096	094	094	094	094
	.6	120	120	112	113	116	116	116	116
				Normal (N = 300)					
.1	.1	055	055	056	054	044	044	044	044
	.3	056	056	056	054	044	044	044	044
	.6	059	059	059	059	049	049	049	049
.4	.1	056	056	052	050	056	056	056	056
	.3	057	057	055	054	052	052	052	052
	.6	054	054	054	053	050	050	050	050
.7	.1	158	158	042	042	054	054	054	054
	.3	082	082	046	047	054	054	054	054
	.6	060	060	054	054	043	043	043	043

(table continues)

TABLE 2. (Continued)

Pred–crit. corr.	Pred. int.	H	WH	O	M	WS	D	S1	S2
				Uniform (N = 300)					
.1	.1	046	046	047	046	056	056	056	056
	.3	044	044	046	044	058	058	058	058
	.6	046	046	047	046	048	048	048	048
.4	.1	052	052	046	045	040	041	040	040
	.3	047	047	042	042	052	052	052	052
	.6	044	044	044	044	036	036	036	036
.7	.1	142	142	044	044	047	047	047	047
	.3	071	071	038	038	042	042	042	042
	.6	037	037	033	033	026	027	027	026
				Exponential (N = 300)					
.1	.1	049	049	050	048	048	049	048	048
	.3	052	052	053	051	048	048	048	048
	.6	048	048	050	048	050	050	048	048
.4	.1	070	070	066	064	072	072	072	072
	.3	070	070	070	068	071	071	071	071
	.6	072	072	070	070	064	064	064	064
.7	.1	171	171	063	064	068	068	068	068
	.3	132	132	091	091	089	089	089	089
	.6	128	128	122	123	109	109	109	109

Note. Decimals have been omitted for clarity of presentation. Pred–crit. corr. = predictor–criterion correlation. Pred. int. = predictor intercorrelation. Predictor–criterion correlation refers to the magnitudes of ρ_{yx1} and ρ_{yx2}. H = Hotelling's *t*. WH = Williams's modified *t* per Hendrickson, Stanley, and Hills (1970). O = Olkin's *z*. M = Meng, Rosenthal, and Rubin's *z*. WS =Williams's (1959) "standard" *t*. D = Dunn and Clark's *z*. S1 = Steiger's original modification of Dunn and Clark's *z*. S2 = Steiger's modification of Dunn and Clark's *z* using a backtransformed *z* procedure. The alpha estimates for Hotelling's *t*, Williams's modified *t* per Hendrickson, Stanley, and Hills (1970), Olkin's *z*, and Meng, Rosenthal, and Rubin's *z* were originally published in May and Hittner (1997).

I error when sampling from a normal distribution, there were six instances in which the Type I error rate was somewhat liberal (i.e., ranging from .058 to .063). One other Type I error rate trend within the normal distribution is worth noting. In particular, for the larger sample sizes (i.e., 50 and above) the remaining four of the eight statistical tests (i.e., Williams's standard *t*, Dunn and Clark's *z*, and Steiger's two modifications of Dunn and Clark's *z*) yielded Type I error rates that hovered around the nominal level.

For the uniform distribution, six of the eight statistical tests (with the exceptions of Hotelling's *t* and Hendrickson et al.'s modification of Williams's *t*) exhibited conservativeness as the predictor–criterion correlation increased. Furthermore, for the largest predictor–criterion correlation (.7), the tests became more conservative as the predictor intercorrelation increased. Finally, the exponential distribution showed an opposite trend. As shown in Table 2, the Type I error rates for the six tests increased as the predictor–criterion correlation increased. Moreover, for the largest predictor–criterion correlation (.7), the tests yielded dramatically higher Type I error rates compared with the nominal level.

Statistical Power

The simulated power estimates for all eight statistical tests are presented in Table 3. As the data indicate, the power simulations revealed a similar pattern of findings across the three population distributions (normal, uniform, and exponential). For the smallest sample size ($N = 20$), we obtained acceptable levels of power (approximately .80 and higher) for most cases with an effect size of .6 regardless of the predictor intercorrelation. For the sample size of $N = 50$ with normal and uniform population distributions, we obtained acceptable levels of power for all cases with an effect size of .6 and for most cases with an effect size of .3 (with the exception of predictor intercorrelations of .1). For the exponential distribution (a highly skewed distribution), the findings for $N = 50$ were similar, except that for an effect size of .3 and a predictor intercorrelation of .3, four of the estimates were clearly unacceptable (power = .65) and the remaining four estimates were borderline (ranging from .72 to .74). For all three population distributions and across all three levels of predictor intercorrelation, the results for $N = 100$ yielded acceptable levels of power for effect sizes of .3 and .6. We obtained similar results for the largest sample size ($N = 300$), except that consistently acceptable levels of power also emerged when effect sizes were .1 and predictor intercorrelations were .6.

Several trends emerged with regard to differences in estimated power across the statistical tests. First, there was a general tendency across many situations for Hotelling's *t* and Williams's *t* per Hendrickson et al. to yield the highest power estimates. That result was understandable, given the aforementioned higher Type I error estimates, and was consistent with the contention by Steiger (1980) that Hotelling's test yields inflated power estimates at the expense of Type I error rate (Steiger did not examine Hendrickson and colleagues' version of Williams's *t*). In some cases, the degree to which power estimates were comparatively large for Hotelling's *t* and Hendrickson et al.'s version of Williams's *t* was quite pronounced. For example, under the normal distribution condition with $N = 300$, effect size = .1, and predictor intercorrelation = .1, the power estimates for those two tests were .80, whereas the remaining six statistics yielded estimates of either .67 or .68. In some cases under very small sample size conditions ($N = 20$), Olkin's *z* test likewise resulted in higher power estimates that were also accompanied by inflated Type I error rates.

TABLE 3. Power Estimates for all Eight Statistics

ES	Pred. int.	H	WH	O	M	WS	D	S1	S2
				Normal (N = 20)					
.1	.1	.22	.22	.13	.12	.12	.13	.13	.13
	.3	.18	.18	.14	.14	.12	.13	.13	.12
	.6	.18	.18	.17	.17	.15	.16	.15	.15
.3	.1	.44	.44	.40	.38	.36	.38	.37	.37
	.3	.45	.45	.44	.41	.41	.42	.41	.41
	.6	.58	.58	.59	.57	.60	.61	.60	.60
.6	.1	.79	.79	.81	.77	.76	.77	.76	.76
	.3	.88	.88	.89	.87	.87	.88	.87	.87
	.6	.99	.99	.99	.99	.99	.99	.98	.98
				Uniform (N = 20)					
.1	.1	.21	.21	.12	.12	.11	.11	.11	.11
	.3	.15	.15	.12	.11	.10	.11	.11	.11
	.6	.15	.15	.14	.14	.14	.15	.14	.14
.3	.1	.43	.43	.39	.36	.36	.37	.37	.37
	.3	.45	.45	.45	.42	.38	.40	.39	.38
	.6	.61	.61	.62	.59	.58	.59	.58	.58
.6	.1	.82	.82	.83	.80	.78	.79	.78	.78
	.3	.89	.89	.90	.88	.88	.89	.88	.88
	.6	.99	.99	.99	.99	.99	.99	.99	.99
				Exponential (N = 20)					
.1	.1	.26	.26	.18	.18	.16	.17	.17	.17
	.3	.22	.22	.19	.19	.19	.20	.19	.19
	.6	.23	.23	.22	.22	.20	.21	.21	.20
.3	.1	.46	.46	.43	.40	.38	.40	.39	.39
	.3	.47	.47	.46	.44	.42	.44	.43	.43
	.6	.61	.61	.61	.59	.58	.59	.58	.57
.6	.1	.76	.76	.78	.74	.74	.76	.75	.74
	.3	.84	.84	.86	.83	.82	.83	.82	.82
	.6	.97	.97	.97	.97	.97	.97	.97	.96
				Normal (N = 50)					
.1	.1	.33	.33	.21	.21	.20	.21	.21	.21
	.3	.28	.28	.24	.24	.23	.23	.23	.23
	.6	.31	.31	.30	.30	.30	.30	.30	.30
.3	.1	.75	.75	.70	.69	.68	.69	.68	.68

(table continues)

TABLE 3. (Continued)

ES	Pred. int.	H	WH	O	M	WS	D	S1	S2
				Normal (N = 50)					
.3	.3	.79	.79	.78	.77	.76	.76	.76	.76
	.6	.92	.92	.92	.92	.93	.93	.93	.93
.6	.1	.98	.98	.98	.98	.98	.99	.99	.99
	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
				Uniform (N = 50)					
.1	.1	.32	.32	.21	.20	.20	.20	.20	.20
	.3	.26	.26	.21	.21	.21	.22	.22	.22
	.6	.29	.29	.28	.27	.27	.28	.27	.27
.3	.1	.77	.77	.71	.70	.71	.71	.71	.71
	.3	.81	.81	.79	.78	.78	.78	.78	.78
	.6	.93	.93	.93	.93	.94	.94	.94	.94
.6	.1	.99	.99	.99	.99	.99	.99	.99	.99
	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
				Exponential (N = 50)					
.1	.1	.36	.36	.26	.26	.25	.26	.26	.26
	.3	.32	.32	.29	.29	.30	.30	.30	.30
	.6	.36	.36	.35	.35	.33	.34	.33	.33
.3	.1	.71	.71	.68	.66	.65	.65	.65	.65
	.3	.74	.74	.73	.72	.65	.65	.65	.65
	.6	.88	.88	.88	.88	.90	.90	.90	.90
.6	.1	.96	.96	.96	.95	.97	.97	.97	.97
	.3	.99	.99	.99	.99	.99	.99	.99	.99
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
				Normal (N = 100)					
.1	.1	.47	.47	.34	.33	.31	.31	.31	.31
	.3	.43	.43	.36	.36	.37	.37	.37	.37
	.6	.48	.49	.47	.47	.47	.47	.47	.47
.3	.1	.94	.94	.91	.91	.93	.93	.93	.93
	.3	.97	.97	.96	.96	.95	.95	.95	.95
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
.6	.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

(table continues)

TABLE 3. (Continued)

ES	Pred. int.	H	WH	O	M	WS	D	S1	S2
				Normal (N = 100)					
.6	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
				Uniform (N = 100)					
.1	.1	.47	.47	.32	.32	.32	.32	.32	.32
	.3	.42	.42	.35	.35	.35	.35	.35	.35
	.6	.50	.50	.48	.48	.48	.48	.48	.48
.3	.1	.95	.95	.94	.94	.94	.94	.94	.94
	.3	.98	.98	.97	.97	.97	.97	.97	.97
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
.6	.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
				Exponential (N = 100)					
.1	.1	.48	.48	.36	.36	.36	.36	.36	.36
	.3	.44	.44	.39	.39	.39	.39	.39	.39
	.6	.50	.50	.49	.49	.48	.48	.48	.48
.3	.1	.91	.91	.89	.88	.88	.88	.88	.88
	.3	.92	.92	.92	.92	.92	.92	.92	.92
	.6	.99	.99	.99	.99	.98	.98	.98	.98
.6	.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
				Normal (N = 300)					
.1	.1	.80	.80	.67	.67	.68	.68	.68	.68
	.3	.78	.78	.73	.73	.72	.72	.72	.72
	.6	.88	.88	.87	.87	.86	.86	.86	.86
.3	.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
.6	.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

(table continues)

TABLE 3. (Continued)

ES	Pred. int.	H	WH	O	M	WS	D	S1	S2
				Uniform (N = 300)					
.1	.1	.79	.79	.66	.66	.65	.65	.65	.65
	.3	.78	.78	.74	.74	.74	.74	.74	.74
	.6	.90	.90	.90	.90	.89	.89	.89	.89
.3	.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
.6	.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
				Exponential (N = 300)					
.1	.1	.77	.77	.64	.64	.64	.64	.64	.64
	.3	.73	.73	.68	.68	.66	.66	.66	.66
	.6	.82	.82	.81	.81	.79	.79	.79	.79
.3	.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
.6	.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	.6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Note. ES = effect size. ES refers to the magnitude of difference between the two predictor–criterion correlations, $\rho_{y.x1}$ and $\rho_{y.x2}$. The specified correlations used to generate each effect size are as follows: (a) for ES = .1, ρ's = .7 and .6, (b) for ES = .3, ρ's = .7 and .4, and (c) for ES = .6, ρ's = .7 and .1. Pred. int. = predictor intercorrelation. H = Hotelling's *t*. WH = Williams's modified *t* per Hendrickson, Stanley, and Hills (1970). O = Olkin's *z*. M = Meng, Rosenthal, and Rubin's *z*. WS = Williams's (1959) "standard" *t*. D = Dunn and Clark's *z*. S1 = Steiger's original modification of Dunn and Clark's *z*. S2 = Steiger's modification of Dunn and Clark's *z* using a backtransformed *z* procedure. The power estimates for Hotelling's *t*, Williams's modified *t* per Hendrickson, Stanley, and Hills (1970), Olkin's *z*, and Meng, Rosenthal, and Rubin's *z* were originally published in May and Hittner (1997).

In general, the other four procedures being considered (i.e., Williams's standard *t*, Dunn and Clark's *z*, and Steiger's two modifications of Dunn and Clark's *z*) were essentially equivalent with regard to power. The only difference was that Dunn and Clark's *z* had a minute increase in power (.01) in certain situations with small sample sizes. That negligible difference dissipated with increasing sample size.

Discussion

In the present study, we extended and refined previous simulation research in two important ways. First, we used the methodology of May and Hittner (1997) to factorially cross the variables of sample size, population distribution, magnitude of predictor intercorrelation, effect size or magnitude of discrepancy between $\rho_{y,x1}$ and $\rho_{y,x2}$ (for power), and magnitude of the predictor–criterion correlations (for Type I error rate) to create the conditions for generating empirical estimates of power and Type I error rate. Because that approach generates estimates of power and Type I error rate for a multitude of 3×3 correlation matrices (i.e., correlations among x_1, x_2, and y), the resultant estimates should be of greater use to applied researchers.

A second way in which the present study builds on previous research is that we examined a relatively large number of statistical tests for comparing dependent correlations and we included two tests that have not yet been evaluated via Monte Carlo simulation in the peer-reviewed literature (i.e., Steiger's two modifications of Dunn and Clark's z). Given the number of test statistics we examined and the multitude of situations for which empirical estimates were generated, the present study yielded a plethora of findings, some of which suggest that conclusions based on previous research should be tempered.

With regard to the Type I error rate findings, our data support previous assertions (e.g., Steiger, 1980) that Hotelling's t demonstrates inflated power at the expense of Type I error rate. Specifically, Hotelling's t (and Hendrickson et al.'s modification of Williams's t) yielded higher Type I error rates when the predictor–criterion correlation was .7 and the predictor intercorrelations were .1 and .3. In a similar fashion, under certain conditions (i.e., predictor–criterion correlations of .1 and .4, small sample size, normal and uniform population distributions), Olkin's z also demonstrated relatively higher Type I error rates than did the other test statistics. Given the inflated Type I error rates associated with those three tests and considering that the other statistics that we examined performed consistently better within the normal and uniform population distributions, we cannot recommend that applied researchers use Hotelling's t, Hendrickson et al.'s modification of Williams's t, or Olkin's z.

With regard to the remaining statistical tests, four of the five procedures (i.e., Williams's standard t, Dunn and Clark's z, and Steiger's two modifications of Dunn and Clark's z) were approximately equivalent in their ability to control the Type I error rate. However, for the smallest sample size ($N = 20$), Dunn and Clark's z yielded Type I errors that were closer to the nominal level (i.e., less conservative). The final statistical test (Meng et al.'s z test) controlled Type I error rate fairly effectively but yielded several liberal empirical alpha estimates for a normal population distribution.

Of additional interest in the normal and uniform population distributions was the finding that Steiger's original modification of Dunn and Clark's z did not offer more Type I error rate control than that provided by Dunn and Clark's z. Apparently, the process of averaging or pooling correlations before the calculation of the covariance term in Dunn and Clark's formula does not improve Type I error rate control (at least not in the simulated conditions examined in the present study). Steiger's modification of Dunn and Clark's z using a backtransformed average Fisher's z procedure did not control Type I error rate any more effectively than did either Dunn and Clark's z or Steiger's (original) modification of Dunn and Clark's z. It might be that because Dunn and Clark's z and Steiger's modification of Dunn and Clark's z already demonstrate good overall statistical properties, the incorporation of a Fisher's z backtransformation procedure was unable to further improve Type I error rate control.

Finally, the finding that all of the Type I error rates exceeded $\alpha = .05$ for the exponential distribution when the predictor–criterion correlations were .4 and .7 suggests that researchers who work with highly skewed distributions might first want to normalize their data with the use of either the Box–Cox transformation (Box & Cox, 1964), which is the optimum procedure, or one of the noniterative normalizing transformations (e.g., square root, logarithmic, inverse) before they calculate correlation coefficients. The transformed data will then meet the normality assumption required of Pearson's r (to the extent that the normalizing transformations are successful), and the various tests for comparing dependent correlations can then be applied. However, one should note that a normalizing transformation might not be necessary when the predictor–criterion correlations are small (.1). Under such conditions for the exponential distribution, our simulation findings indicated that most of the tests for comparing dependent correlations maintain good Type I error rate control (see Table 2).

Our findings indicated that previous assertions in the literature about the low power of most statistical tests under small sample size conditions need to be qualified. In particular, our data indicated that when sample size was small ($N = 20$) and effect size was large (.6), all eight statistical tests demonstrate adequate levels of power. Similarly, all eight statistics demonstrated adequate levels of power when sample size was 50 and effect size was either moderate (.3) or large (.6). The only exception for $N = 50$ occurred under moderate effect size conditions when the predictor intercorrelation was small (.1). Another statement from the literature that requires modification is the assertion that high levels of power are always attainable when sample size is large (i.e., Ns of 300 or more). Our findings indicated that the extent to which that statement is true largely depends on the values of both the effect size and the predictor intercorrelation. In fact, our results indicated that when sample size was large ($N = 300$) and effect size was small (.1), the levels of power are uniformly acceptable only when the predictor intercorrelation is large (.6). In other words, for $N = 300$, power is inadequate for most of the test statistics when the effect size is small (.1) and the predictor intercorrelation is small or moderate (.1 or .3). As those findings indicate. the power of statistical tests for comparing correlated correlations depends not only on sample size and population distribution but also on the magnitude of discrepancy between $\rho_{y,x1}$ and $\rho_{y,x2}$ (effect size) and the degree of predictor intercorrelation ($\rho_{x1,x2}$).

The fact that Hotelling's t demonstrated consistently higher levels of power than did six of the remaining seven tests is consistent with the assertion by Steiger (1980) that Hotelling's t yields inflated power at the expense of Type I error rate. The finding that Hendrickson et al.'s modification of Williams's t likewise yielded inflated power at the expense of Type I error rate, and did so to approximately the same degree as Hotelling's t, is also noteworthy and might be a topic worth investigating more thoroughly in future research. A related finding that also might be worth investigating is the fact that for some situations in which $N = 20$, Olkin's z test also yielded inflated power estimates at the expense of Type I error rate. We cannot recommend that applied researchers use Hotelling's t, Hendrickson et al.'s modification of Williams's t, or Olkin's z when they test the difference between dependent correlations, given the findings on inflated power and Type I error rate and the fact that viable alternatives are available (see hereinafter).

The remaining five statistics were all quite comparable in their Type I error rate control and statistical power. However, because there were several occasions for the normal distribution in which Meng et al.'s z test yielded inflated alpha estimates, we are inclined to recommend that researchers use one of the remaining four statistical procedures. Of the remaining four procedures, it appears that Williams's (1959) standard t and Dunn and Clark's (1969) z have the best overall statistical properties. Our finding that Williams's t effectively controlled Type I error rate for the normal and uniform distributions is consistent with previous simulation research (e.g., Boyer et al., 1983; Neill & Dunn, 1975; Steiger, 1980) that has also found Williams's t to perform well under a variety of conditions.[1] We also recommend Dunn and Clark's z on the basis of our findings that (a) neither of the two modifications controlled Type I error rate better than did Dunn and Clark's z and (b) Dunn and Clark's z demonstrated a minute increase in power over Williams's t and the two modifications in certain situations with small sample sizes.

In closing, our simulation findings extend previous research on the robustness of Pearson's r to violations of population normality (Havlicek & Peterson, 1977; Zeller & Levine, 1974), and they build on past work that has examined the robustness of tests for comparing dependent correlations (Boyer et al., 1983; May & Hittner, 1997; Neill & Dunn, 1975). In future simulation studies on dependent correlations, researchers might consider the examination of a broader array of nonnormal, as well as mixed-normal, population distributions. Edgell and Noon (1984), for example, found that the t test for Pearson's r, which is used to test the null hypothesis that $\rho = 0$, was robust under a variety of mixed-normal conditions. Similarly, Yu and Dunn (1982) found that an asymptotic z test for comparing two dependent correlations maintained good Type I error rate control for mixed-normal distributions. An important direction for future research is to extend the work of Yu and Dunn by evaluating the extent to which the tests that were examined in the present study would show robustness under a variety of mixed-normal population distributions. Finally, we believe that there are two major conclusions to be drawn from our simulation results. First, given the inflated Type I error rates that Hotelling's t, Hendrickson et al.'s modification of Williams's t, and Olkin's z demonstrated in certain situations, we cannot recommend that applied researchers use those procedures. Second, when Type I error rate and statistical power are considered simultaneously, it appears that Williams's standard t and Dunn and Clark's z maintained the best overall statistical properties. Thus, we recommend that applied researchers use one of the two aforementioned procedures when they test the significance of difference between dependent correlation coefficients.

NOTE

1. It should be noted, however, that consistent with Boyer et al. (1983), we also found that the performance of Williams's t was compromised when a highly skewed population distribution was used. In particular, Williams's t and all seven of the remaining statistics that we examined yielded inflated Type I error rates for the exponential distribution when the predictor–criterion correlations were .4 and .7.

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